

공학수학 I (BS102a-01)

학기	2026년 1학기
담당교수	안흥주
이수구분	공통필수
학점	3.0
강의시간	화13:00-15:00(E7 - 223), 목13:00-15:00(E7 - 223)
강의실	
소속	기초학부

교과개요

"The book of the universe is written in the language of mathematics." Galileo wrote this four hundred years ago, even before Newton and Leibniz discovered calculus. The statement is as valid today as ever: We use functions in all the sciences, and calculus allows us to analyze the functions and draw scientific conclusions. After a quick and brief review of differentiation and integration, we will study the differential equation, more precisely, the ordinary differential equation (ODE).

The derivative measures the instantaneous rate of change of a function. The definite integral measures the total accumulation of a function over an interval. These two ideas form the basis for nearly all mathematical formulas in science. The rules by which we can compute the derivative (respectively, the integral) of any function are called calculus. The fundamental theorem of calculus links the two processes of differentiation and integration beautifully.

We also cover topics related to complex numbers and matrices, which are frequently encountered in science and engineering texts.

교과목표

Course Objectives

- Can understand and apply basic concepts related to differential equations.
- Understands analytical, qualitative, and numerical solutions of the first and second order differential equations
- Can understand and apply various properties related to the differentiation and integration of single-variable functions
- Can understand and apply the fundamental theorem of calculus.
- Can understand and apply the concepts of sequence limits, infinite series, and Taylor's theorem.
- Can understand and apply the concepts of parametric representation of curves and polar coordinates.
- Understands the solution of differential equations using the Laplace transform

주차별 강의계획

1주차:See the attached file

2주차:

3주차:

4주차:

5주차:

6주차:

7주차:

8주차:
9주차:
10주차:
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15주차:
16주차:

학습윤리

0. Attendance is the most basic behavior that confirms the sincerity of learning activities, and attendance verification requires students to be honest. Unethical behavior in attendance is strictly prohibited, such as proxy attendance and unauthorized early departure.

1. Writing and submitting assignments is the act of presenting the results of one's learning, which can be enhanced by including the learner's own thoughts and expressions in the assignment. Unethical behavior in writing and submitting assignments includes falsification and alteration, plagiarism, and duplicate submissions, which are strictly prohibited.

2. Cooperative learning is the act of acquiring diverse knowledge and creating new knowledge and should always be based on your own sincere efforts. Free-riding is the act of taking advantage of the hard work of others without putting in a fair amount of effort, which is unethical because it harms the people involved.

3. Tests are designed to measure students' academic performance and are based on fair assessment. Cheating can be done by asking someone else to take the test or by using someone else as a substitute.

교과정책(방침)

Course Policies

□Must attend recitation (연습시간). Recitation consists of working through Matlab, solving exercises, asking questions, etc. and may be facilitated by a TA

□Up to 5 unexcused absences allowed. More than 6 unexcused absences (20%) may result in a lower grade than the original.

□Same as above for attendance for recitation, but allow up to 3 absences (20% excluding exam week)

□Absolute evaluation in the class (절대평가)

□Failure to attend midterm and final exams will result in an F grade regardless of score (there is no make-up exam)

□Anything below 20 points out of 100 (20 is not an absolute score but with some variation depending on the tests) is an F.

□In the event of an unavoidable withdrawal by the student, credit may reasonably be granted, taking into account the points earned by the student, as evidenced by clear documentation

□

□반드시 연습시간을 수강 해야함. 연습시간에는 매트랩 실습, 연습문제 풀이, 학생의 질문 등으로 이루어지며, 실습 등을 수행할 때 조교가 진행할 수 있음

□질의 및 면담 시간: 강의시간에 별도 공지함. TA 면담 시간은 연습 시 간에 별도 공지

□5번까지 결석 허용. 6회 이상(20 %) 정당한 사유 없이 무단 결석한 경우, 본래보다 낮은 학점이 부여될 수 있음. (강의 시작 15분 이후 입실할 경우 지각으로 정의하고. 지각 3번은 결석 1번으로 정의하며. 결석에는 경조사 및 천재지변 등을 모두 포함함. 5회 미만일 경우 따로 교수에게 알릴 필요 없음)

□연습 시간에 대한 출석도 위와 동일하게 적용하되 결석 회수는 3회까지 허용(시험 주간 제외20%)

□05 반 학생만을 대상으로 절대 평가.

□중간, 기말 고사 중 하나를 치르지 않으면 점수와 관계없이 F 처리 됨.

□100점 만점 환산 20점(절대적 점수는 아니고 시험의 난이도에 따라 약간의 차이 있음)미만은 F

□학생이 피할 수 없는 유고 치 못할 경우 명확한 서류에 의하여 증명되고 담당교수가 인정하는 한에서 학생이 취득한 점수를 고려하여 합리적으로 점수를 부여할 수 있음.

□매주 일정한 분량의 숙제(HW)가 주어지고, 숙제는 정해진 제출기한까지 LMS 과제물 게시판 또는 지정된 장소에

제출.

□교과 내용과 Matlab 교육 내용을 각각 2번의 퀴즈를 통해 테스트함. 교과 퀴즈 및 Matlab 퀴즈는 아래 강의 계획서에 계획된 일자에 진행하며 원칙적으로 모든 시험은(퀴즈 포함) 재시험 없음.